

Lab - Hard Drive Maintenance

Introduction

In this lab, you will examine the results of using Disk Check and Disk Defragmenter on a hard drive.

Recommended Equipment

- A computer running Windows
- Two or more partitions on the hard drive

Instructions

Part 1: Run the Error-checking Tool on a disk volume.

- Log on to Windows as an administrator.
- Open the **File Explorer** > click **This PC**.
If there is only one partition on the hard drive, use **Disk Management** to create a second partition with at least 2GB.
- Right-click the volume without an installed Windows OS. Click **Properties**.
- In the Properties window, click the **Tools** tab. Click **Check** to start the error checking.
- Windows monitors the drives and runs automatic scheduled scans. You may see a message saying that **You don't need to scan this drive**. Click **Scan drive**.
- When you receive a message that your drive or disk was successfully scanned, click **Show details**.
- The **Event Viewer** opens to display the **Chkdsk** event. Double click the **Information** event in the middle pane to review the detailed information.

How many stages were processed?

Were any problems found with the volume? If problems were found, what were they?

- Click **Close**. Close the **Event Viewer** as necessary.

Part 2: Use the Disk defragmenter tool.

Note: Do not perform this step if your computer has an SSD drive. It is unnecessary to defragment SSD drives.

- Navigate to the **Properties** window for the drive that does not have a Windows OS installation.
- In the Properties window, click the **Tools** tab. Click **Optimize** to start disk optimization.
- Select the volume without an installed Windows OS. Click **Analyze** in the Optimize Disk window.

What percentage of the disk is fragmented?

- Click **Optimize** to start defragmenting the disk.

What is the first process during defragmenting (See **Current status** column)?

What are the three tasks performed for each pass (See **Current status** column)?

How many passes did it take to defragment drive?

- e. When defragmenting is complete, click **Close**.
- f. Close all windows.

Note: It is not possible to view the detail of the defragmented hard drive through the GUI version of defragmenter.